

is used. For setting this protocol, the `pthread_mutexattr_setprotocol` function is used and the value of the protocol is `PTHREAD_PRIO_INHERIT`. When a thread blocks a higher priority thread owing to a mutex, with `PTHREAD_PRIO_INHERIT` it inherits the priority of the higher priority thread being blocked and executes with that priority. Once the execution is over, it goes back to normal priority. In such a case, M cannot pre-empt L; hence, L completes its execution with boosted priority. Once it releases the mutex lock, H starts executing and, finally, when H finishes off, M starts executing.

Priority inversion is a problematic scenario that is more common in single-core processor systems. In the case of multi-core processor systems, this problem is avoided because of the execution of threads on different cores. Multiple solutions exist for this problem such as priority ceiling, random boosting, etc. We have used the priority inheritance protocol to avoid priority inversion. Pthread mutex locks not only provide a synchronisation mechanism, but also provide support for priority inheritance. **END** 

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```
E:\>cd Python27
E:\Python27>cd blockchain
E:\Python27\blockchain>python blockchainserver.py
* Serving Flask app "blockchainserver" (lazy loading)
* Environment: production
WARNING: Do not use the development server in a production environment.
Use a production WSGI server instead.
* Debug mode: off
* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
```

Figure 4: Performing the transaction using cURL

```
E:\curl-7.65.0-win64-mingw\bin>curl localhost:5000/mine
{"timestamp": "2019-05-25 20:05:42.338000", "data": {"transactions": [{"to": "Receiver", "amount": 5, "from": "Sender"}, {"to": "Person-2", "amount": 5, "from": "Person-1"}, {"to": "Person-1", "amount": 1, "from": "network"}], "proof-of-work": 18, "hash": "8ae9acf42e4c4b89384818b0bd1a1ce48640734fcc5f2813bfb4368eed47d4d", "idx": 1}}
```

Figure 5: Mining the records and transactions

blockchain using a Web server so that there is distributed deployment.

```
$ curl "http://localhost:5000/myblockchain" -d '{"from": "ss", "to": "fsd", "amount": 3}' -H "Content-Type: application/json"
```

Using cURL, the transaction can be implemented and its impact on the blockchain can be visualised. The cURL library for the Windows OS can be installed from <https://curl.haxx.se/windows/>.

As depicted in Figure 6, the execution of the code and the overall implementation with all the records and transactions can be analysed so that there is transparency in the operations, preventing any hacking attempts. Using Proof of Work

```
* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
127.0.0.1 - - [25/May/2019 20:03:36] "POST /txion HTTP/1.1" 404 -
127.0.0.1 - - [25/May/2019 20:04:01] "POST /txion HTTP/1.1" 404 -
New transaction
Sender: Sender
Receiver: Receiver
Amount: 5
127.0.0.1 - - [25/May/2019 20:04:23] "POST /myblockchain HTTP/1.1" 200 -
New transaction
Sender: Person-1
Receiver: Person-2
Amount: 5
```

Figure 6: Recording all transactions on the server

(PoW), the integrity of transactions is logged and committed.

Scope for research and development

In the current scenario, governments as well as corporate organisations are striving towards implementing blockchain technology for secured applications. For such integrations, there is a need to associate the secured algorithms of Proof of Work (PoW) to ensure the privacy and integrity in the implementations. Research scholars and forensic scientists can make use of blockchain technologies to make accurate predictions about specific identities, for forensic as well as law enforcement scenarios. **END** 

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